

BasicSynth User Manual

AudioForge BasicSynth v1.0.0

Subtractive synthesizer with 3 oscillators, ADSR envelope, and resonant filter

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Overview

BasicSynth is a classic subtractive synthesizer featuring:

- **3 Oscillators:** Sine, sawtooth, and square waveforms
- **ADSR Envelope:** Attack, Decay, Sustain, Release
- **Low-pass Filter:** Resonant filter with cutoff and Q controls
- **Polyphony:** Up to 8 voices
- **MIDI:** Full note and velocity support

- **Simple & CPU-efficient:** Perfect for learning and production

When to Use BasicSynth

- **Bass lines:** Deep, punchy bass with sawtooth + filter
 - **Lead synth:** Bright leads with square wave + envelope
 - **Pads:** Soft, evolving pads with sine wave + slow attack
 - **Sound design learning:** Understand subtractive synthesis fundamentals
 - **CPU-light production:** When you need synth sounds without heavy CPU
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Quick Start

Play Your First Note

1. **Load BasicSynth** as an instrument in your DAW
2. **Connect a MIDI keyboard** or use your DAW's piano roll
3. **Play a note** (e.g., C3)
4. **Hear sound!** (default: sawtooth wave, bright and buzzy)

Create a Bass Sound (30 seconds)

1. **Enable Sawtooth oscillator** (OSC 2 button)
2. **Disable other oscillators** (OSC 1, OSC 3)
3. **Lower filter cutoff** to 500-800 Hz
4. **Increase filter resonance** to 30-50%
5. **Shorten envelope release** to 0.1-0.2s
6. Result: **Punchy, filtered bass**

Create a Pad Sound (30 seconds)

1. **Enable Sine oscillator** (OSC 1 button)

2. **Disable other oscillators**
3. **Increase attack** to 1.0-2.0s
4. **Increase release** to 2.0-3.0s
5. **Open filter cutoff** to 2000-5000 Hz
6. Result: **Soft, swelling pad**

User Interface

OSCILLATORS			
☐	☐	☐	
OSC1	OSC2	OSC3	← On/Off
SINE	SAW	SQR	← Waveform
☐	☐	☐	
50%	50%	50%	← Level (0-100%)
FILTER			
☐	☐		
○	○	← Cutoff, Resonance	
☐	☐		
Cutoff	Resonance		
1000 Hz	20%		
ENVELOPE (ADSR)			
☐	☐	☐	☐
○	○	○	○
☐	☐	☐	☐
Attack	Decay	Sustain	Release
0.01s	0.1s	70%	0.3s
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Parameters

Oscillators (OSC 1, 2, 3)

OSC 1 - Sine Wave:

- **Waveform:** Pure sine wave (fundamental only)
- **Character:** Smooth, soft, no harmonics
- **Use:** Sub-bass, pads, sine-bass

OSC 2 - Sawtooth Wave:

- **Waveform:** Sawtooth (all harmonics, decreasing amplitude)
- **Character:** Bright, buzzy, rich harmonics
- **Use:** Leads, bass, brass-like sounds

OSC 3 - Square Wave:

- **Waveform:** Square (odd harmonics only)
- **Character:** Hollow, clarinet-like, nasal
- **Use:** Leads, chiptune sounds, retro game music

Level Control (0-100% for each):

- Adjusts volume of each oscillator
- Mix multiple oscillators for layered sounds
- 50% default = balanced mix

Filter

Cutoff (20 Hz - 20 kHz):

- **Low values** (200-800 Hz): Dark, muffled, bass-heavy
- **Mid values** (1-5 kHz): Warm, balanced
- **High values** (5-20 kHz): Bright, open, full harmonics

Resonance (0-100%):

- **0%**: No resonance (smooth rolloff)
- **20-40%**: Adds character, slight peak at cutoff
- **60-80%**: Strong resonance, whistling sound
- **90-100%**: Self-oscillation (filter resonates on its own)

Filter Type: 12 dB/octave low-pass (Moog-style)

Envelope (ADSR)

Attack (0.001s - 5.0s):

- Time for sound to go from silence to full volume
- **Fast (0.01-0.1s)**: Plucky, percussive sounds (bass, leads)
- **Slow (1.0-5.0s)**: Swelling pads, ambient sounds

Decay (0.001s - 5.0s):

- Time to fall from attack peak to sustain level
- **Fast (0.01-0.1s)**: Snappy, defined (piano-like)
- **Slow (0.5-2.0s)**: Smooth transition (strings)

Sustain (0-100%):

- Level held while key is pressed
- **0%**: Sound decays to silence (percussive)
- **100%**: Held at full volume (organ-like)
- **50-80%**: Natural decay (most instruments)

Release (0.001s - 5.0s):

- Time to fade to silence after key release
 - **Fast (0.01-0.2s)**: Abrupt stop (staccato)
 - **Slow (1.0-5.0s)**: Long tail (reverb-like)
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Sound Design Guide

Bass Sounds

Sub Bass (deep, low-end power):

OSC 1 (Sine): 100%
OSC 2 (Saw): 0%
OSC 3 (Square): 0%
Filter Cutoff: 200-400 Hz
Filter Res: 10-20%
Attack: 0.01s, Decay: 0.2s, Sustain: 50%, Release: 0.2s

Reese Bass (modulated, wobbling):

OSC 1 (Sine): 40%
OSC 2 (Saw): 60%
OSC 3 (Square): 0%
Filter Cutoff: 500-1000 Hz (automate for wobble)
Filter Res: 40-60%
Attack: 0.01s, Decay: 0.3s, Sustain: 70%, Release: 0.3s

Pluck Bass (short, percussive):

OSC 2 (Saw): 80%
OSC 3 (Square): 20%
Filter Cutoff: 1500 Hz
Filter Res: 30%
Attack: 0.005s, Decay: 0.15s, Sustain: 0%, Release: 0.1s

Lead Sounds

Bright Lead (cutting through the mix):

OSC 2 (Saw): 60%
OSC 3 (Square): 40%
Filter Cutoff: 4000-8000 Hz
Filter Res: 40%
Attack: 0.01s, Decay: 0.2s, Sustain: 80%, Release: 0.5s

Detuned Lead (two instances, detune by +5 cents):

Instance 1: OSC 2 (Saw) 100%
Instance 2: OSC 2 (Saw) 100%, pitch +5 cents
Both: Filter Cutoff 5000 Hz, Res 30%

Pad Sounds

Soft Pad:

OSC 1 (Sine): 100%
Filter Cutoff: 3000 Hz
Filter Res: 10%
Attack: 1.5s, Decay: 1.0s, Sustain: 60%, Release: 2.5s

Lush Pad (layered):

OSC 1 (Sine): 50%
OSC 2 (Saw): 30%
OSC 3 (Square): 20%
Filter Cutoff: 2500 Hz
Filter Res: 20%
Attack: 2.0s, Decay: 1.5s, Sustain: 70%, Release: 3.0s

Other Sounds

Pluck/Mallet:

OSC 2 (Saw): 70%
OSC 3 (Square): 30%
Filter Cutoff: 3000 Hz
Filter Res: 50%
Attack: 0.005s, Decay: 0.3s, Sustain: 0%, Release: 0.2s

Organ:

OSC 3 (Square): 100%
Filter Cutoff: 8000 Hz (wide open)
Filter Res: 0%
Attack: 0.01s, Decay: 0.05s, Sustain: 100%, Release: 0.05s

Presets & Techniques

Technique 1: Layer Multiple Instances

Goal: Create rich, complex sounds

How:

1. Insert 2-3 instances of BasicSynth on the same track
2. **Instance 1:** Sub bass (sine, low cutoff)
3. **Instance 2:** Mid harmonics (saw, mid cutoff)
4. **Instance 3:** High air (square, high cutoff)

Result: Full-spectrum bass or lead

Technique 2: Filter Automation

Goal: Add movement and expression

How:

1. Create a MIDI clip with held notes
2. **Automate filter cutoff** from low (200 Hz) to high (5000 Hz)
3. **Automate filter resonance** from 20% to 60%

Result: Classic filter sweep (dubstep wobble, disco filter)

Technique 3: Velocity → Filter Cutoff

Goal: Harder notes = brighter sound (expressive)

How:

1. In your DAW, map MIDI velocity to filter cutoff
2. Soft notes (velocity 40-60) = dark, mellow
3. Hard notes (velocity 100-127) = bright, aggressive

Note: BasicSynth v1.0 doesn't have built-in velocity routing - use DAW's MIDI CC mapping.

Technique 4: Combine with Effects

BasicSynth alone is simple - add effects for pro sounds:

Bass:

- BasicSynth (sub bass) → **Saturation** (harmonics) → **Reverb** (space)

Lead:

- BasicSynth (bright lead) → **Chorus** (width) → **CleanDelay** (depth)

Pad:

- BasicSynth (soft pad) → **Reverb** (long tail) → **SimpleComp** (glue)

MIDI Implementation

Note Range

- **C0 to C8** (full MIDI range)
- **Polyphony**: 8 voices (notes beyond 8 will steal oldest voice)

Velocity Sensitivity

- **Velocity** → **Volume**: Harder notes = louder (0-127 MIDI velocity)
- **Velocity** → **Filter**: Not implemented in v1.0 (use DAW MIDI CC)

MIDI CC Support

BasicSynth responds to these MIDI CC messages:

- **CC 1 (Mod Wheel)**: Not assigned (future: vibrato)
- **CC 7 (Volume)**: Overall synth volume
- **CC 10 (Pan)**: Stereo pan
- **CC 11 (Expression)**: Volume (similar to CC 7)

Tip: Map DAW MIDI CC to parameters for real-time control.

Technical Specifications

Synthesis Engine

- **Type**: Subtractive (oscillator → filter → envelope)
- **Polyphony**: 8 voices
- **Voice stealing**: Yes (oldest note)
- **Oscillators**: 3 (sine, saw, square)
- **Filter**: 12 dB/oct low-pass, resonant

- **Envelope:** ADSR (applies to both amplitude and filter)

Performance

- **CPU Usage:** ~1-2% per voice (8 voices = ~8-16% total)
- **Latency:** 0 samples (real-time processing)

Audio Quality

- **Bit Depth:** 32-bit float
- **Sample Rate:** 44.1 - 192 kHz
- **Anti-aliasing:** Yes (on sawtooth and square)

Troubleshooting

No sound when playing notes

Fix:

1. Check at least one oscillator is enabled
2. Verify MIDI is reaching the plugin (check DAW MIDI monitor)
3. Ensure sustain level >0% (or notes will decay to silence)

Sound cuts off abruptly

Cause: Release time too short

Fix: Increase Release to 0.3s or higher

Filter self-oscillates (whistles)

Cause: Resonance too high (>90%)

Fix: Reduce resonance to 60-80% for normal use

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